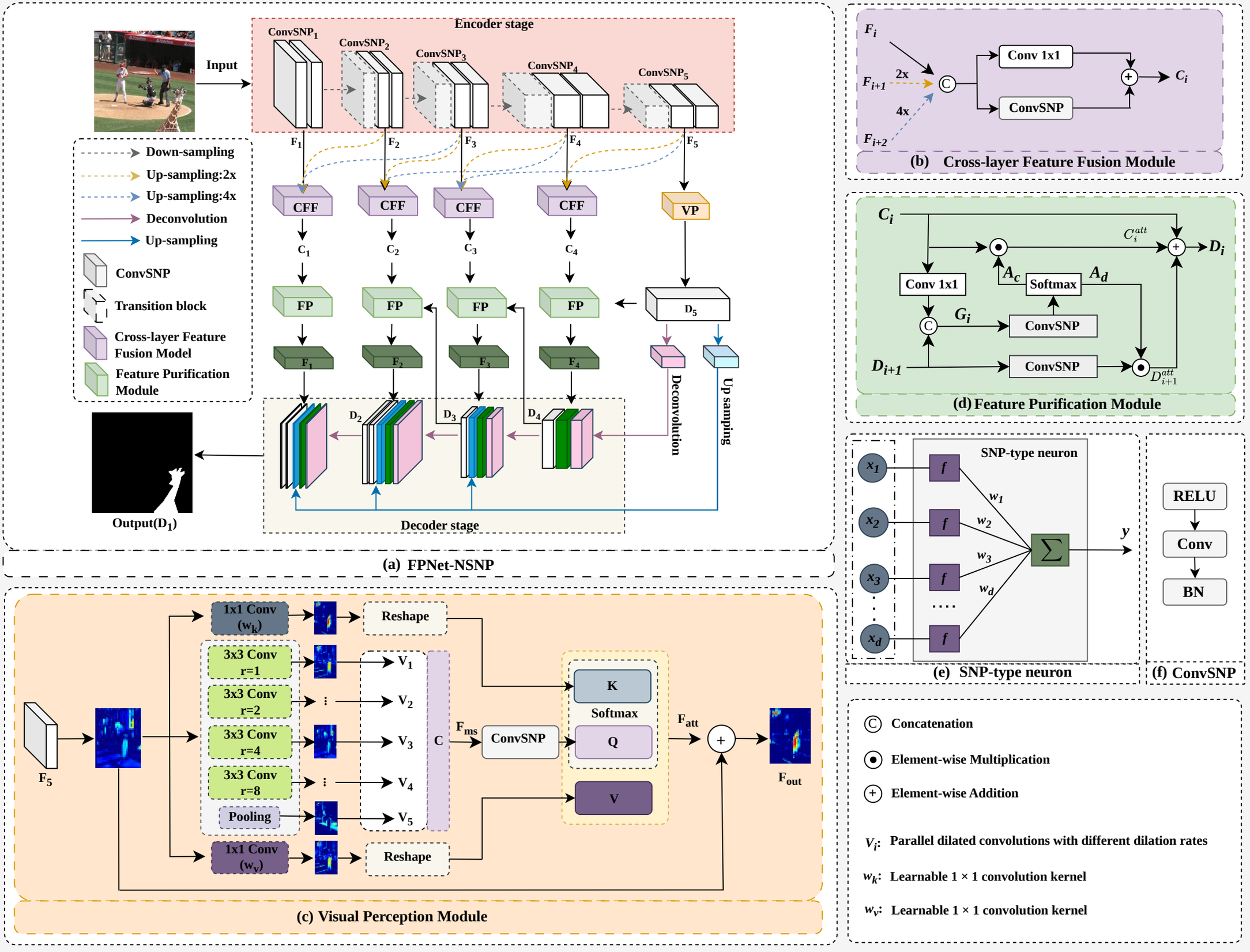

Input
Encoder stage
ConvSNP1
ConvSNP2
ConvSNP3
ConvSNP4
ConvSNP5
F1
F2
F3
F4
F5
CFF
VP
C1
C2
C3
C4
FP
D5
D2
D3
D4
Deconvolution
Up samping
Decoder stage
Output(D1)
Down-sampling
Up-sampling:2x
Up-sampling:4x
Deconvolution
Up-sampling
ConvSNP
Transition block
Cross-layer Feature Fusion Model
Feature Purification Module
(a) FPNet-NSNP
Fi
Fi+1
Fi+2
2x
4x
C
Conv 1x1
ConvSNP
Ci
(b) Cross-layer Feature Fusion Module
Ci
Conv 1x1
Softmax
Ac
Ad
Gi
ConvSNP
Di+1
Di
(d) Feature Purification Module
SNP-type neuron
x1
x2
x3
xd
f
w1
w2
w3
wd
y
(e) SNP-type neuron
RELU
Conv
BN
(f) ConvSNP
1x1 Conv (wk)
Reshape
3x3 Conv r=1
3x3 Conv r=2
3x3 Conv r=4
3x3 Conv r=8
Pooling
V1
V2
V3
V4
V5
C
Fms
ConvSNP
K
Q
V
Softmax
Fatt
Fout
1x1 Conv (wv)
(c) Visual Perception Module
Concatenation
Element-wise Multiplication
Element-wise Addition
Vi: Parallel dilated convolutions with different dilation rates
wk: Learnable 1 × 1 convolution kernel
wv: Learnable 1 × 1 convolution kernel